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Earth and Planetary Science Letters 155 (1998) 1–14

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Paleomagnetic confirmation of the Laurentian origin of the Argentine Precordillera

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Received 23 July 1997; revised 10 October 1997; accepted 28 October 1997

Abstract

Several recent tectonic models have portrayed the Argentine Precordillera has been recently portrayed as an Early Paleozoic Laurentian derived exotic terrane now found in southwestern South America. These models have primarily been based on strong biogeographic and stratigraphic evidence, however, no paleomagnetic data have previously been available to independently test them. A paleomagnetic study was, therefore, carried out on the Early Cambrian Cerro Totora Formation, exposed in the northern reaches of the Argentine Precordillera. After stepwise thermal demagnetization a pre-folding remanence was identified in ten sites of this formation, yielding a paleomagnetic pole (CT) at 37.0°N, 314.1°E, $A_{95} = 5.8°$. This pole is not consistent with the latest Proterozoic–Early Paleozoic apparent polar wander path for Gondwana, but it agrees with the Early Cambrian Section of the Laurentian path if the Argentine Precordillera is positioned as the conjugate margin of the Ouachita embayment in southeast Laurentia. This result confirms that the Precordillera is an allochthonous terrane derived from Laurentia in Cambrian times, that was later accreted to Gondwana, probably in Middle Ordovician times. © 1998 Elsevier Science B.V.

Keywords: paleomagnetism; Precordillera; Paleozoic; Argentina; Gondwana; Laurentia

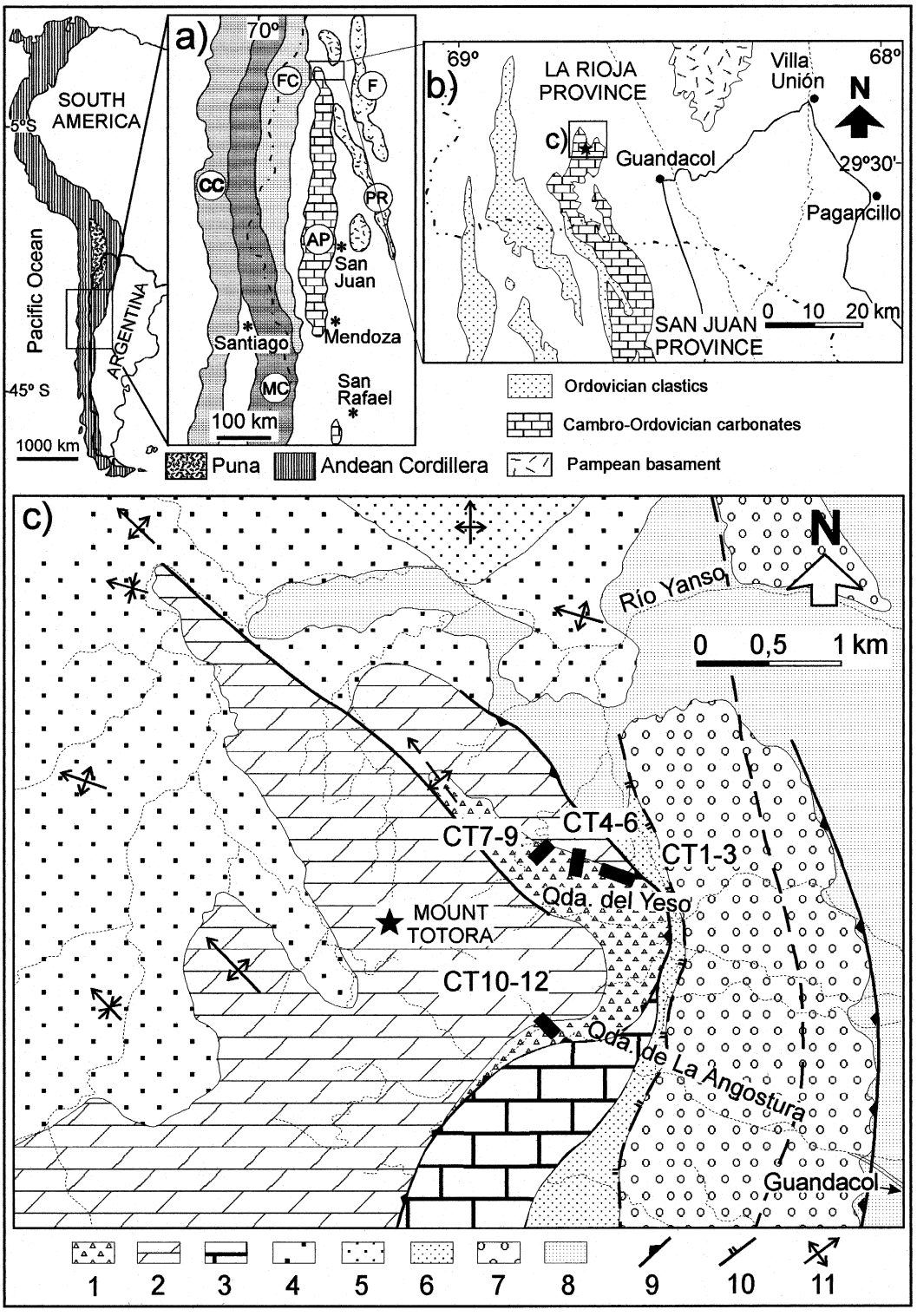
1. Introduction

The Argentine Precordillera (AP), in the foothills of the Argentine Andes (Fig. 1a), has recently been proposed to be an exotic fragment derived from Laurentia because it contains Cambrian fossil faunas and sedimentary successions that are nearly identical to those of continental-shelf successions of Laurentia [2–6]. Isotopic data from the basement rocks of the Argentine Precordillera have also shown much closer affinities with Appalachian basement rocks than with adjacent basement rocks in South America [7]. On this basis, it has been proposed that the Precordillera continental fragment was most probably rifted from the Ouachita embayment of the southern margin of Laurentia in Cambrian times [8] and, after drifting in the Iapetus ocean, was finally accreted to the western

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PII S0012-821X(97)00196-9


a)
70°
SOUTH AMERICA
5°S
Pacific Ocean
ARGENTINA
45° S
1000 km
FC
F
CC
PR
AP
San Juan
Santiago
Mendoza
MC
San Rafael
100 km
Puna
Andean Cordillera
b)
69°
68°
LA RIOJA PROVINCE
Villa Unión
N
c)
Guandacol
29°30'
Pagancillo
0 10 20 km
SAN JUAN PROVINCE
Ordovician clastics
Cambro-Ordovician carbonates
Pampean basament
c)
N
Río Yanso
0 0,5 1 km
CT7-9
CT4-6
CT1-3
Qda. del Yeso
MOUNT TOTORA
CT10-12
Qda. de La Angostura
Guandacol
1 2 3 4 5 6 7 8 9 10 11

margin of Gondwana — the present western side of South America [5,6,9]. This accretion apparently occurred sometime between the Early and latest Ordovician, by which time there is conclusive evidence in both the Precordillera stratigraphy [5,6] and faunal affinities [4,10] to establish a clear connection of the AP and Gondwana. At present, most authors agree on the Laurentian origin of the AP (see [11]). However, as the original rifted margins of both the Appalachian–Ouachita and the AP have been deformed by later orogenies and are, at present, partly covered by younger rocks, the precise rift site and geometry, as well as the paired rifted-margin paleogeography, is difficult to unravel. Hence, alternative interpretations for the relative timing of separation from Laurentia and of accretion to Gondwana, as well as for the exact positioning and kinematics of rifting have been suggested [5,6,8,12–16].

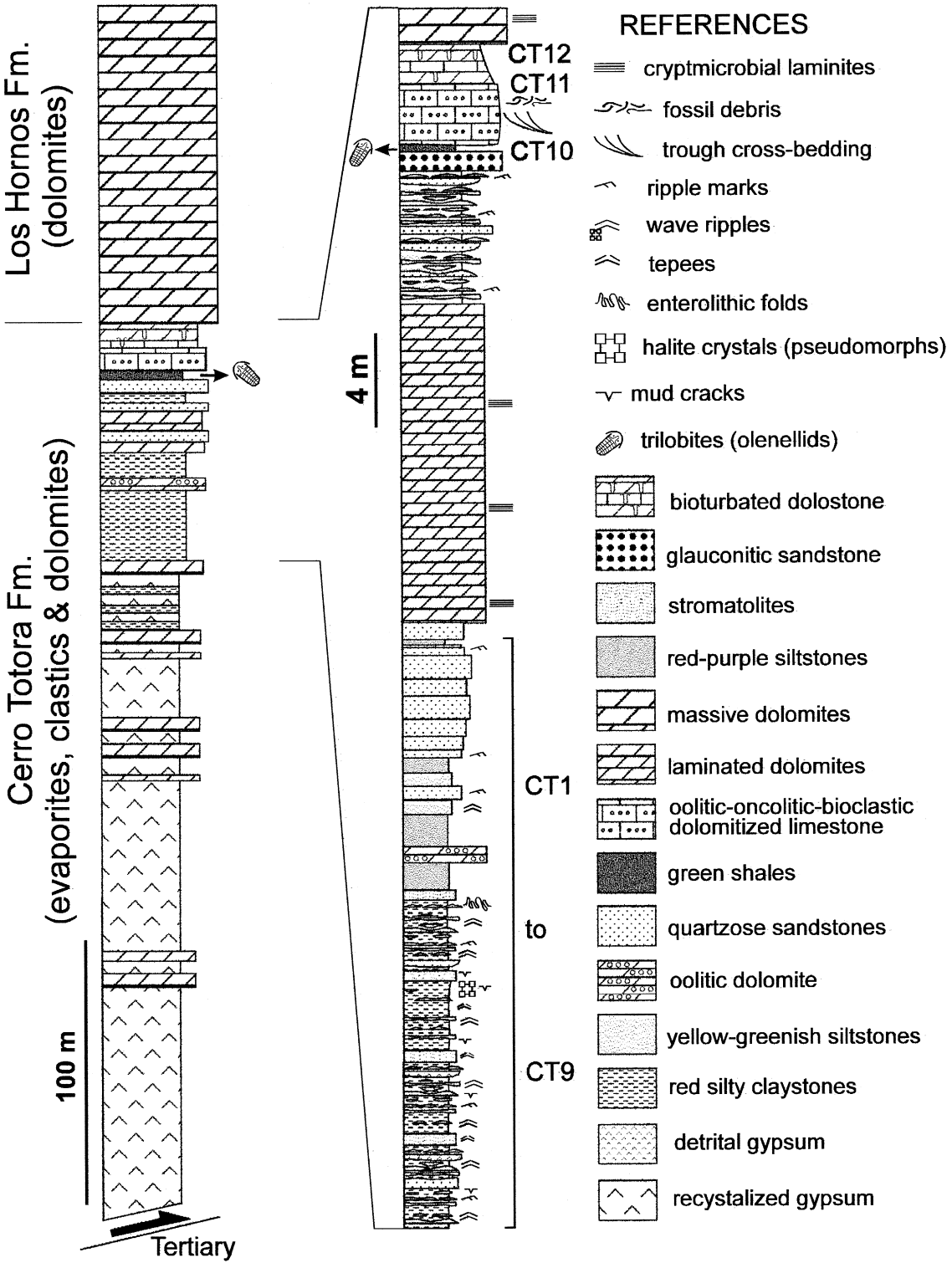
Little paleomagnetic work has been carried out in the Lower Paleozoic rocks of the AP to date. Rapalini and Tarling [17] determined that many of the Lower Paleozoic rocks of western-central Argentina were remagnetized by a major tectonic event, known as the San Rafaelic phase, during the Early Permian. This widespread remagnetization event affected most of the AP as well as areas several hundred kilometers south of it [[17–19]], inhibiting determinations of the primary magnetic signatures. This includes the Late Ordovician Alcaparrosa Formation [20] which has an anomalous pole position, due to its Early Permian age of magnetization, that was mistakenly interpreted as evidence for a possible Laurentian origin of AP [21]. Only a few ill-defined, possibly primary magnetization components were found in latest Cambrian carbonates, but they are inadequate for serious tectonic consideration [17]. Thus this paper contributes the first clear paleomagnetic data that reliably confirm the Laurentian derivation of AP and its probable origin within the Ouachita embayment.

2. Geologic background

The Argentine Precordillera (Fig. 1) consists of a Grenvillian age lithospheric block [7], with a thick cover of folded Paleozoic and Tertiary rocks [22]. According to its morphostructural features it is considered as part of the extensive thrust-fold belt located to the east of the main Andes (Fig. 1a) and is bounded to the east by the Sierras Pampeanas, which have a metamorphic basement of Late Precambrian–Early Cambrian age [23,24]. The Precordilleran thrust-fold belt extends over 400 km in length between the provinces of Mendoza and La Rioja, but the present-day features are strongly controlled by the Andean tectonic regime [25]. Thus, the real north–south length of the exotic block may exceed 800 km when considering its southerly prolongation into what is known as the San Rafael block in southern Mendoza [5].

The most outstanding stratigraphic feature of its Paleozoic evolution is the development of a thick Cambrian to Lower Ordovician carbonate succession [26,27], which strongly contrasts with the synchronous clastic and volcaniclastic successions in the rest of the South American margin. This, together with the Laurentian affinity of the trilobite faunas, led Poulsen [28] and Borrello [2,29] to consider its probable exotic origin (see also [30]). More recently, analyses of brachiopod paleobiogeography led Benedetto [4] to suggest that AP had remained close to Laurentia until the earliest Ordovician. Detailed taxonomic studies then allowed Vaccari (see fig. 5 in [5], and [31]) to establish a correlation with the southeastern margin of Laurentia, with a close match between the northern Precordillera and the southern

Fig. 1. (a) Location map of the Argentine Precordillera with enlargements and geology of the northern region (b) and the sampled locality (c), modified from [1]. *CC* = Coastal Cordillera; *MC* = Main Cordillera; *FC* = Frontal Cordillera; *AP* = Argentine Precordillera, *PR* = Pampean Ranges; *F* = Famatina. *1* = Lower Cambrian evaporites and red clastics of the Cerro Totora Formation; *2* = Lower–Middle Cambrian dolomites of the Los Hornos Formation; *3* = Upper Cambrian–Lower Ordovician dolomites and limestones of the La Flecha–La Silla–San Juan Formations; *4* = Carboniferous–Permian continental arkoses; *5* = Upper Permian–Triassic conglomerates and coarse sandstones; *6* = Tertiary fine-grained red beds; *7* = Plio-Pleistocene synorogenic conglomerates; *8* = Quaternary alluvium; *9* = thrusts; *10* = normal faults; *11* = folds. *Solid rectangles*: location of sampling sites.


Los Hornos Fm.
(dolomites)
Cerro Totora Fm.
(evaporites, clastics & dolomites)
100 m
Tertiary
4 m
CT12
CT11
CT10
CT1
to
CT9
REFERENCES
cryptmicrobial laminites
fossil debris
trough cross-bedding
ripple marks
wave ripples
tepees
enterolithic folds
halite crystals (pseudomorphs)
mud cracks
trilobites (olenellids)
bioturbated dolostone
glauconitic sandstone
stromatolites
red-purple siltstones
massive dolomites
laminated dolomites
oolitic-oncolitic-bioclastic dolomitized limestone
green shales
quartzose sandstones
oolitic dolomite
yellow-greenish siltstones
red silty claystones
detrital gypsum
recystalized gypsum

Appalachians also having been proposed on stratigraphic grounds [32]. This paleomagnetic study was carried out in order to test these tectonic models.

3. Study area, setting and age

The paleomagnetic sampling was carried out in the northern reaches of the classical Precordillera domain in the La Rioja Province (Fig. 1b). Recently, the local stratigraphy for the area was reinterpreted [1] and a new Lower Cambrian unit, the Cerro Totora Formation, was defined. This unit comprises a mixed succession of evaporites, red siliciclastics and carbonates (Fig. 1c and 2), covered by thick Cambrian to Lower Ordovician carbonates (> 2500 m). The Cerro Totora Formation records the transition from synrift evaporites and marginal-marine red and variegated arkosic sedimentary rocks, to quartz arenites and intercalated olive green and glauconitic shales. Such a stratigraphic development allows this unit to be considered as a typical synrift succession [1], reflecting a generalized transgression associated with the cessation of rifting. *Bonnia–Olenellus* Zone trilobites [33] in the green shales and dolomitized grainstones of the upper part of the unit (Fig. 2) suggest an Early Cambrian age for this rifting.

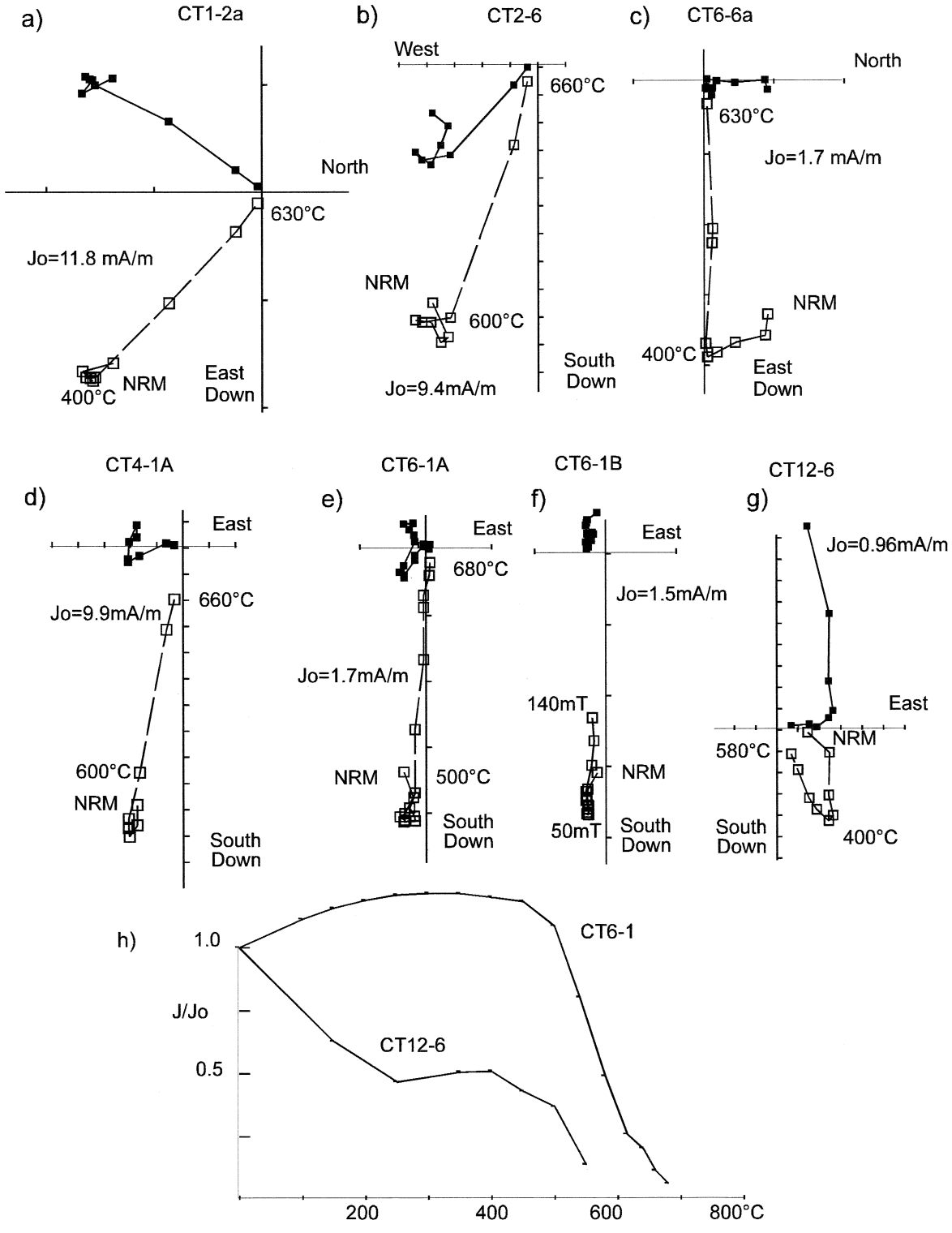
The basal anhydrite–gypsum succession (> 250 m) is a common restricted circulation facies in the early stages of evolving divergent margins in low-latitude arid-climatic settings [34,35]. The evaporites are strongly recrystallized and interbedded with dolomitized cryptomicrobial and oolitic tabular limestones, that are succeeded by a predominantly siliciclastic interval (~ 50 m) composed of red to purplish silty-shales and minor quartzose to subfeldspathic sandstones with sparse evaporite and carbonate layers. This red-bedded interval was the main sampled interval and includes halite hopper-crystal casts as well as intrasedimentary gypsum crystals, which indicate extensive evaporation in arid, supratidal environments, as shown by modern analogs [36,37]. Mud-cracked horizons, together with tepee structures and brecciated horizons, are also common features suggesting subaerially exposed mud-flat and salt-flat environments [38]. In the Late Early Cambrian the onset of a passive-margin stage is recorded by the upward transition to a thick carbonate bank, controlled by thermally-driven tectonic subsidence and eustatic sea-level fluctuations (cf. [39]).

Astini et al. [32] pointed out the time parallelism and the nearly identical lithology of the Cerro Totora Formation with the contemporaneous stratigraphy of the Rome Formation in the southern Appalachians and the Black Warrior Basin, where the rifting history has developed similar graben-fill successions (e.g., Mississippi trough and Birmingham graben, [40,41]). On this basis, Thomas and Astini [8] firmly suggested that the AP should be considered as a block of Laurentian crust rifted from the Ouachita embayment, initially bounded by the Blue Ridge rift to the east, the Alabama–Oklahoma transform to the north and the Ouachita rift to the west. Hence, the AP would have been the southernmost continuation of the Appalachians and formed the conjugate margin of the Texas promontory.

4. Paleomagnetic results

Eighty samples (69 cores and 11 hand-samples) were collected at 12 sites from the Cerro Totora Fm. (Figs. 1 and 2). Nine sites (CT1–CT9) were located in the red siltstone Section, two (CT11 and CT12) on the dolomitized grainstone and one (CT-10) in the green shales located immediately below the grainstones. Whenever possible, samples were oriented with both sun and magnetic compasses. Two or three cores were subsequently drilled from each hand-sample and one or two 2.2 cm high specimens were obtained from each core. Pilot stepwise AF demagnetization in 15 steps up to fields of 140 mT proved to be adequate to identify the magnetic components (Fig. 3f). However, the characteristic component could not be completely erased by this technique as generally over 50% of its original intensity remained

Fig. 2. Stratigraphic log of the Cerro Totora Formation (Lower Cambrian of the Argentine Precordillera) and location of the interval sampled (modified from [1]).


a)
CT1-2a
North
630°C
Jo=11.8 mA/m
NRM
400°C
East
Down
b)
CT2-6
West
660°C
NRM
600°C
South
Down
Jo=9.4mA/m
c)
CT6-6a
North
630°C
Jo=1.7 mA/m
NRM
400°C
East
Down
d)
CT4-1A
East
Jo=9.9mA/m
660°C
600°C
NRM
South
Down
e)
CT6-1A
East
680°C
Jo=1.7mA/m
NRM
500°C
South
Down
f)
CT6-1B
East
Jo=1.5mA/m
140mT
NRM
50mT
South
Down
g)
CT12-6
Jo=0.96mA/m
East
NRM
580°C
South
Down
400°C
h)
1.0
J/Jo
0.5
CT6-1
CT12-6
200
400
600
800°C

after applying a maximum field of 140 mT. High-temperature thermal demagnetization, in fourteen steps up to 680°C, enabled us to identify and define the magnetic components because it permitted a complete deletion of the characteristic remanence. Consequently, thermal demagnetization was applied to the whole collection in nine to twelve stages up to a maximum temperature of 660°C. Typical demagnetization plots of the Cerro Totora Fm. are shown in Fig. 3. Sites CT1–CT9, consisting of red siltstones to very fine-grained sandstones, showed two well-defined magnetic components. A low-temperature component (A) was isolated in many samples at temperatures below 400°C. This component was generally directed northward with moderate to low negative inclinations. Its direction resembles the present-day dipole field in the area. A second component (B) was defined in almost all samples from the red siltstone sites at temperatures higher than 400°C up to 600–660°C. This component was of exclusively steep downward inclinations (in situ coordinates) and generally trended towards the co-ordinate origin. Unblocking temperatures well above 600°C suggest hematite as the main remanence carrier in the siltstones (Fig. 3h). Samples from sites CT-10 to CT-12 showed similar magnetic behaviour, although the unblocking temperatures of the higher-temperature component were always lower than 600°C (Fig. 3h), suggesting than in these three sites the remanence is mainly carried by magnetite (or Ti-poor titanomagnetite). IRM experiments produced composite acquisition curves (Fig. 4), which show a dominant high-coercivity magnetic mineral (hematite) in the red siltstones, and a low-coercivity phase (magnetite) dominating in the dolomites.

Both components, A and B, were defined by means of principal component analysis [42]. In the case of component B, at least four consecutive steps were considered for its definition. Maximum angular deviations (MAD) under 12° were accepted although most (71%) samples showed MAD $< 5°$. B component directions for two samples from site CT-8 could not be defined by these means due to large overlapping of the unblocking temperature spectra between both components. In these two cases great-circle analysis [43] was performed.

Component A directions cluster in geographic coordinates near the present dipole direction for the study area (dec.: 0°, inc.: $-48.5°$), which falls inside the α_{95} of the average of the site mean directions: dec.: 1.2°, inc.: $-41.0°$, α_{95}: 9.3°, $N = 12$, $k = 22.6$ (Fig. 5). Application of the bedding correction to these directions produce a much larger scatter, as shown by the statistical parameters $\alpha_{95} = 13.7°$, $k =$ 11.0. This is a clear indication that the A component is a post-tectonic, recently acquired, secondary magnetization of possible viscous or thermoviscous origin.

Site mean directions for component B (Table 1; Fig. 6) showed an excellent to acceptable within site consistency, as reflected in α_{95} values ranging from 2.4° to 14.8°, except for site CT-11 ($\alpha_{95} = 38°$). This site was therefore excluded from any further analysis. Similarly, the mean direction from site CT-10 is clearly outlier, both in situ and after bedding correction, and this was also excluded from any further consideration. Simple bedding correction was applied by rotating the remanence directions around the strike of the beds the amount of dip at each site. Although no detailed structural analysis of the study area has been carried out yet, plunging of the fold axes towards the northwest (Fig. 1) was found to be gentle and therefore considered not to be a source of significant distortion during bedding correction. This is confirmed by the lack of correlation between the corrected declination values and the structural attitude. Bedding correction produces a substantial improvement in the grouping of the mean site directions; k increases from 15 to 51, while α_{95} is reduced from 12.9° to 6.8° upon 100% tectonic correction. This indicates a positive fold test at a 99% confidence level [44]. Since the folding of the

Fig. 3. Demagnetization plots showing typical magnetic behaviour of samples from the Cerro Totora Formation. (a) to (e) Red siltstone samples during thermal demagnetization; (f) red siltstone sample during AF demagnetization; (g) limestone sample submitted to thermal demagnetization; (h) comparison of normalized demagnetization curves of a red siltstone (hematite bearing) and a limestone (magnetite bearing) sample. *Open* and *solid symbols* correspond to vector representations on the vertical and horizontal plane, respectively.

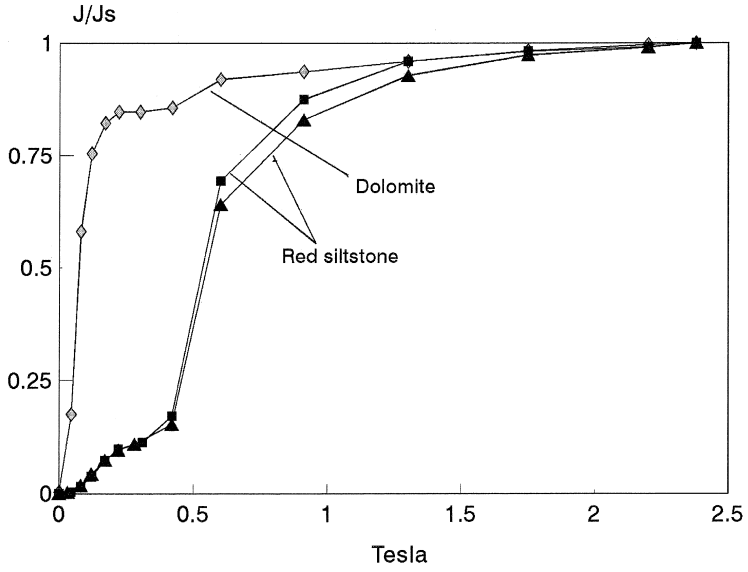


Fig. 4. Normalized isothermal remanent magnetization acquisition curves for three representative samples of the Cerro Totora Formation. See discussion in the text.

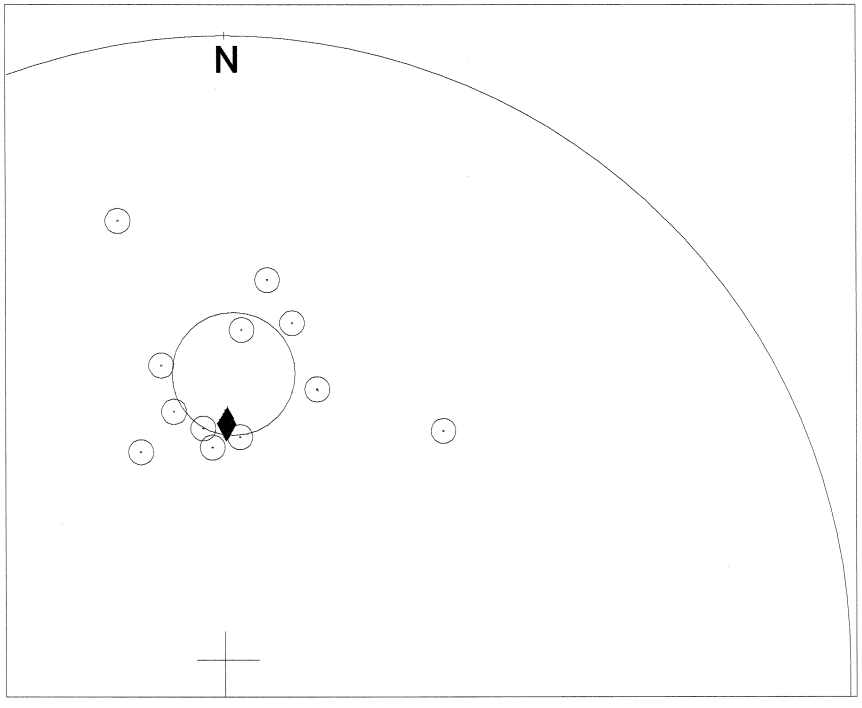


Fig. 5. In situ mean site directions of the A component from the Cerro Totora Formation. The $\alpha 95$ of their average is also shown as a *large circle*. *Solid diamond* indicates present dipole direction in the study location.

Table 1
Mean site characteristic remanence data from the Cerro Totora Formation

Site	n/no.	Dec. in situ (°)	Inc. in situ (°)	α_{95} (°)	Bed. str. (°)	Bed. dip (°)	Dec. corr. (°)	Inc. corr. (°)	VGP lat. (°N)	VGP long. (°E)
CT-1	9/9	232.4	51.4	7.2	307	90	25.0	37.0	34.4	320.0
CT-2	6/6	216.7	73.7	10.3	305	76	34.5	30.2	33.4	331.9
CT-3	8/8	355.3	80.2	6.6	295	55	19.6	26.3	42.7	317.6
CT-4	8/8	232.4	76.0	6.1	289	59	8.7	42.2	35.4	301.0
CT-5	7/7	248.1	75.3	4.7	305	56	23.5	45.8	29.0	315.2
CT-6	6/6	232.9	87.5	2.4	277	57	4.8	34.7	41.2	297.3
CT-7	5/6	63.7	77.3	3.4	273	40	18.1	42.7	33.0	311.0
CT-8[a]	5/7[b]	19.7	61.2	14.7	285	45	17.4	16.3	49.3	316.3
CT-9	6/7[b]	51.6	78.2	5.5	280	45	19.6	35.7	37.2	314.7
CT-10[c]	4/7	152.5	41.1	13.1	269	39	107.1	69.3	−33.2	334.8
CT-12	5/6	83.5	64.9	4.1	256	44	21.5	43.6	31.3	314.1

n = number of samples used in the analysis; no. = number of samples demagnetized; Dec. = mean declination; Inc. = mean inclination; α_{95} = half-angle of the cone of 95% confidence about the mean; Bed. str./dip = strike and dip of bedding, respectively. Mean site directions (in situ): $N = 10$, Dec. = 276.4°, Inc. = 88.8°, $k = 15.1$, $\alpha 95 = 12.9°$; mean site directions (100% correc.): $N = 10$, Dec. = 19.3°, Inc. = 35.8°, $k = 51.2$, $\alpha_{95} = 6.8°$; mean VGP: $N = 10$, lat. = 37.0°N, long. = 314.1°, $A_{95} = 5.8°$.
[a]The mean site direction was computed by the combined analysis of demagnetization circles (2 samples) and principal component determinations (3 samples), following the method of McFadden and McElhinny [43].
[b]Numbers correspond to cores drilled out of five hand samples collected at each site.
[c]Site not considered in computing the paleomagnetic pole.

Cerro Totora Formation in the studied locality is most likely of Late Tertiary age, the positive fold test only indicates a pre-Late Tertiary magnetization. However, it also provides the most reliable paleohorizontal for computing the location of the paleomagnetic pole. A virtual geomagnetic pole was computed

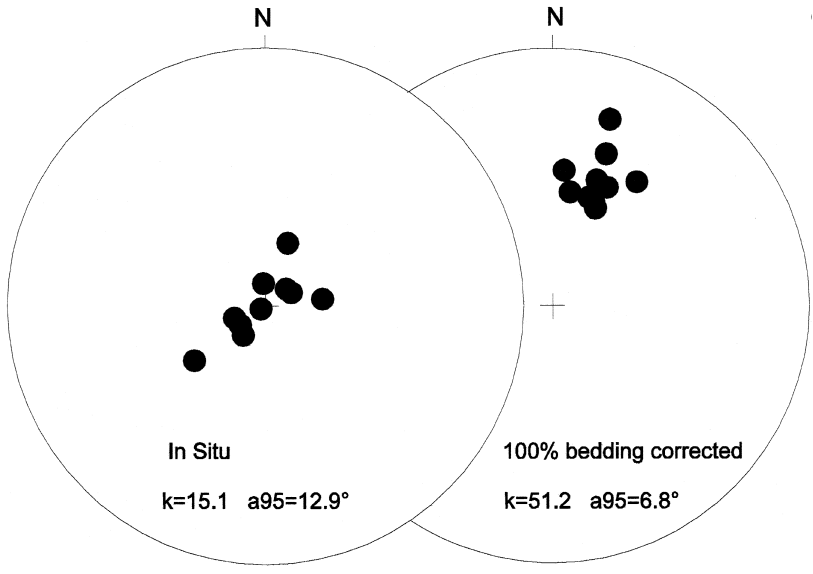


Fig. 6. Mean site directions of component B (characteristic magnetization) from the Cerro Totora Formation, in situ and after 100% bedding correction. The statistical parameter k corresponding to their mean is indicated. Its substantial increase after bedding correction indicates the pre-folding nature of this component. Paleomagnetic data are presented in Table 1.

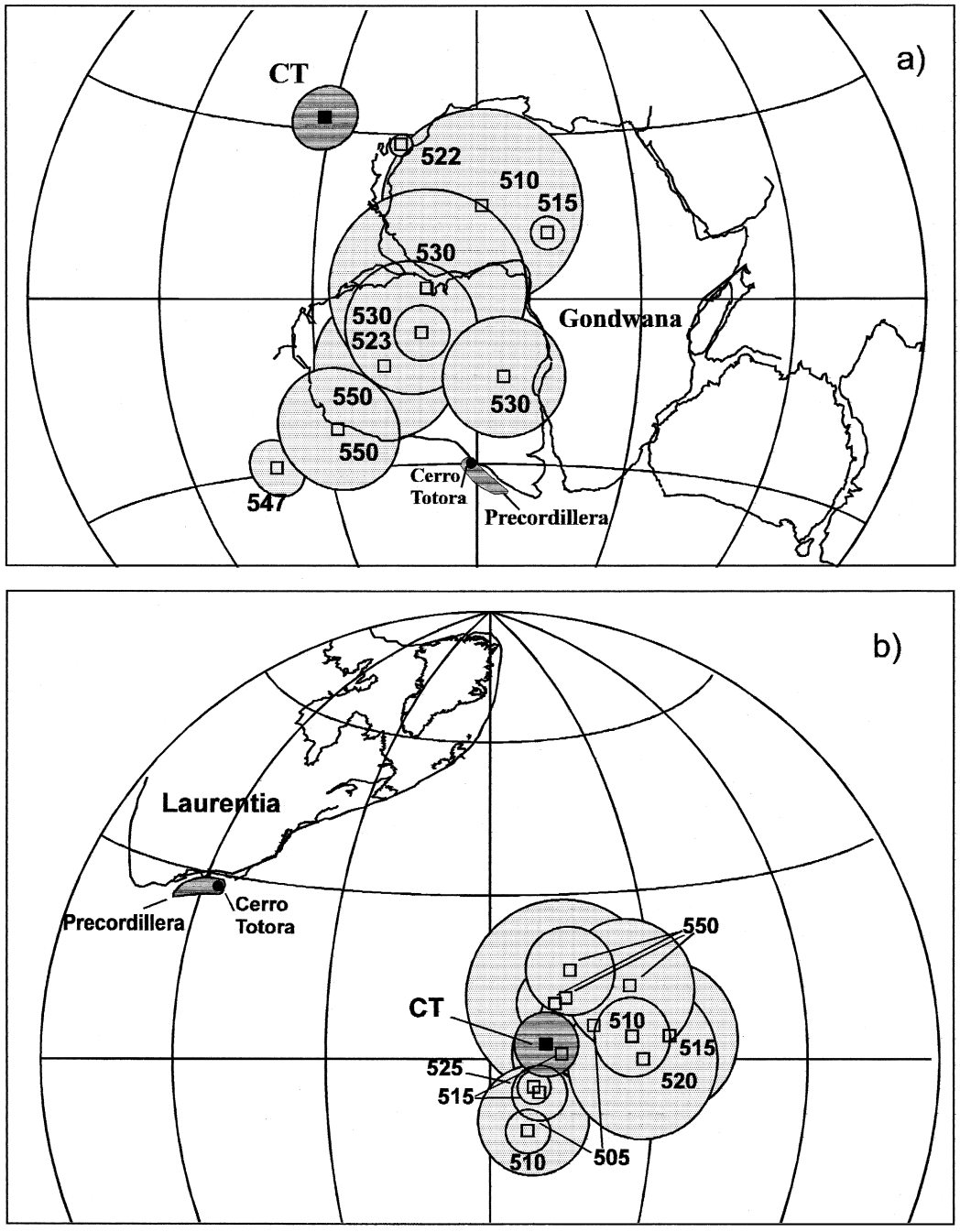


Fig. 7. (a) Cerro Totora Formation paleomagnetic pole (CT) in African coordinates after Gondwana reconstruction [46] and the Gondwana 550–510 Ma APWP [47]. The Precordillera has been kept attached to South America in its present location. *Numbers* indicate the most likely age of the Gondwana poles in Ma [47].
(b) Position of CT after placing the Argentine Precordillera against the southeastern Laurentian margin following the model of Thomas and Astini [8] (rotation parameters: lat.: 8.5°N, long.: 312.1°E, 96.6° clockwise), and the 550–505 APWP of Laurentia [48]. *Numbers* indicate the most likely age of each Laurentian paleomagnetic pole in Ma.

from each site mean direction (Table 1), the mean of which is the paleomagnetic pole for the Cerro Totora Formation: CT, 37.0°N, 314.1°E, $N = 10$, $A_{95} = 5.8°$.

5. Interpretation and discussion

The CT pole position is not consistent with any post-Cambrian expected pole position for Gondwana and/or South America. It seems unlikely, then, that the isolated remanence could be due to remagnetization at a much later age than deposition or early diagenesis (e.g., [45]), unless a very complex scenario of remagnetization and tectonic rotation(s) is invoked. The case for a primary remanent magnetization being determined from the Cerro Totora Formation is supported also by the fact that the same remanence direction is carried by two different minerals (i.e., hematite and magnetite) in two different lithologies (i.e., red siltstones and dolomitic limestones). Inspection of thin section showed that hematite appears in the red siltstones and fine-grained sandstones of this formation as part of the cement, probably as an early diagenetic product. In particular, the analyzed rocks show no evidence of the Permian remagnetization that affects most previously studied Early Paleozoic rocks in the Argentine Precordillera [17,18]. This cannot be attributed entirely to the different lithology of the red siltstone Section (no similar lithology had been previously studied), as one limestone site also shows the primary remanence. The reasons why these rocks escaped remag-

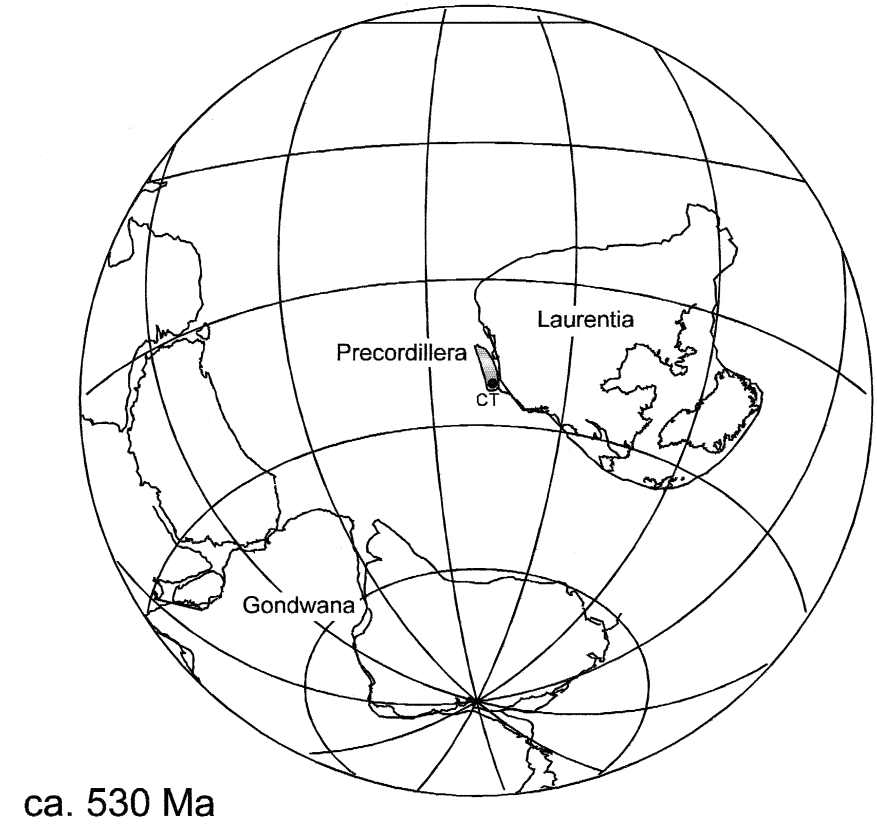


Fig. 8. Paleomagnetically based paleogeographic reconstruction of Laurentia, Precordillera and Gondwana around 530 Ma. Given the paleolongitudinal indetermination of paleomagnetic data the Argentine Precordillera (AP) has been placed against the southeastern margin of Laurentia on the basis of the strong geologic evidence mentioned in the text. Note that the paleolatitude and orientation of the Precordillera determined by this paleomagnetic study perfectly match those from the Ouachita Embayment. The relative paleolongitudinal position of Gondwana is speculative. This reconstruction has been based on the Cerro Totora pole (this paper) for AP, an interpolated pole position between mean poles of 550 and 505 Ma for Laurentia [50], and the mean Australian pole at 530 Ma [51] is used for Gondwana.

netization that affected a large part of the Paleozoic succession of the AP may be related to its more northeasterly location, perhaps far enough from the ''remagnetization front'' of the San Rafaelic Event.

Therefore, we accept an age for the CT pole broadly equal to the geologic age of the Cerro Totora Formation. This is the first Early Paleozoic paleomagnetic pole for the Argentine Precordillera and it can be used to define the paleogeographic position of this terrane in the Early Cambrian.

Fig. 7a shows the latest Proterozoic–Cambrian apparent polar wander path for Gondwana recently presented by Meert and Van der Voo [47], which indicates that the Eastern and Western Gondwana cratonic nuclei were already assembled by ~ 550 Ma. Preliminary results from the Rio de La Plata craton [49] confirm this. Although more high-quality data for the 550–500 Ma Section are needed to refine the path, its main characteristics seem relatively well established. The new CT pole is inconsistent with the Gondwana path if the AP is assumed to have been in its present location with respect to South America. It implies far lower paleolatitudes ($20.1 \pm 5.8°$) than would correspond to the present location of AP as defined by the Early Cambrian (~ 530 Ma) Gondwana poles. Thus the new paleomagnetic data strongly support the Argentine Precordillera as being an Early Paleozoic allochthonous terrane.

When the CT pole is compared with the 550–500 Ma path of Laurentia, by repositioning the AP against the southeastern Laurentian continental margin, according to the hypothesis of Thomas and Astini [8], then it is perfectly consistent with the Laurentian path and corresponds to the most likely position for such a Laurentian pole at ~ 530 Ma (Fig. 7b). This result thus gives very strong additional support to the hypothesis that the AP is an exotic terrane, which rifted from the southeastern continental margin of Laurentia, most likely from the Ouachita embayment.

A paleomagnetically based paleogeographic reconstruction for Early Cambrian times (Fig. 8) shows the probable relative geographic locations of Laurentia, Gondwana and the Argentine Precordillera in the Early Cambrian. The position of the AP respect to Laurentia is nearly identical to that deduced by Thomas and Astini [8] on non-paleomagnetic grounds. The consistency in paleolatitude and orientation of the hypothetical conjugate margins of AP and the Ouachita embayment, obtained from the paleomagnetic data, is remarkable. The longitudinal relative positions between Gondwana and Laurentia–AP are much more speculative.

Further paleomagnetic studies on Middle Cambrian to Middle Ordovician rocks of AP may provide a better constraint on the timing of rifting and accretion, as well as delimiting its intermediate paleogeographic evolution.

6. Conclusions

A paleomagnetic study of the Early Cambrian Cerro Totora Formation, exposed in the northern reaches of the Argentine Precordillera has yielded the first reliable Early Paleozoic paleomagnetic pole for this terrane.

A pre-tectonic, probably primary remanence, carried by hematite or magnetite, was isolated from ten sampling sites. The studied locality, the northeasternmost yet sampled from Early Paleozoic rocks of the Argentine Precordillera, apparently escaped the widespread Early Permian remagnetization that affected many other units of this age in this region.

The paleomagnetic pole is not consistent with the 550–500 Ma path from the already assembled Gondwana. Instead, if the Precordillera is located against the southeastern margin of Laurentia it becomes perfectly consistent with coeval poles from Laurentia.

The paleomagnetic data obtained are a strong evidence that the Argentine Precordillera is an exotic terrane rifted away from the southeastern continental margin of Laurentia.

Further paleomagnetic studies on Middle Cambrian to Middle Ordovician rocks of the AP are necessary to define its geodynamic evolution between its rifting from Laurentia and accretion to Gondwana.

Acknowledgements

This study was supported by a grant of the University of Buenos Aires (UBA-CyT EX135) to AER.

Logistical assistance was provided by CONICOR and the University of Cordoba (Argentina). A. Krittian collaborated during the field work. D.H. Tarling critically reviewed an early draft of the manuscript. The paper benefited from the reviews by Allen McNamara and Conall MacNiocaill. This is a contribution to IGCP Project 376 (Laurentia–Gondwana connections). *[RV]*

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